

Arunakiri Venkatachalam

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PROFESSIONAL SUMMARY

Data Scientist with 8+ years of experience building end-to-end AI and data solutions across the automotive and pharmaceutical industries. Specializes in scalable AI tools, ML pipelines, and On-premises applications. Proven track record of delivering quantified impact: ~60% reduction in document-search time, 30% reduction in third-party software licensing costs, and ~80% OCR accuracy on unstructured regulatory data.

TECHNICAL SKILLS

Languages & Frameworks: Python, Django, REST APIs, SQLAlchemy, Streamlit, Linux Shell

Machine Learning & AI: ML Pipelines, NLP, LLM Agents, RAG Pipelines, LangChain, Vector Databases, MCP

Data Engineering: ETL Pipeline Design, PostgreSQL, Apache Iceberg, AWS S3, High-Frequency Data Processing

DevOps & Cloud: Docker, Kubernetes, CI/CD, Git / GitHub, Keycloak, OAuth 2.0

Analytics & Visualization: Power BI, Grafana, Databricks, Plotly Dash, KPI Dashboard Design

PROFESSIONAL EXPERIENCE

Data Scientist · Volvo Group

October 2023 – Present

- Design and maintain scalable data solutions on secure on-premises servers, Data Science Lab, supporting 200+ users.
- Developed production-ready RAG pipelines using LangChain, enabling natural-language querying of technical documents, reducing document search time by 60%.
- Integrated LLM Agents with MCP (Model Context Protocol) connectors to PostgreSQL and Iceberg tables and render charts directly within the chat interface.
- Developed AI agents for Rigs data analysis using pipeline workflows and tool skills including data retrieval, data conversion, signal filtering, statistical analysis, report generation, visualization, and interactive notebook creation.
- Built high-frequency ML pipelines and interactive Plotly Dash web applications for hydrogen engine test rigs and fleet data streams, reducing pipeline latency by over 40%.
- Built real-time monitoring dashboards in Grafana and Databricks, delivering operational visibility across multiple hydrogen engine test rigs.
- Developed a full-stack access-management portal (Django + PostgreSQL) that automated Data Science Lab user provisioning and resource allocation, eliminating manual ticketing.
- Led the Volvo Data Science Conference 2026, coordinating cross-functional sessions and knowledge-sharing tracks across 20+ teams site-wide.
- Collaborated closely with hardware and software engineers to define data schemas, ingestion strategies, and reporting standards for new engine test programs.

Lead Analyst · HCL Technologies

October 2021 – October 2023

- Delivered end-to-end data engineering solutions as a consultant embedded in Volvo's internal teams, spanning multiple business units and analytical programs.
- Completed my master's Thesis on Volvo's on-premises infrastructure cost-benefit model and predictive forecasting framework, identifying key capital expenditure trends and strategic budget optimization opportunities.
- Member of the MATLAB-to-Python migration initiative, cutting third-party software licensing costs by 30% while improving solution reproducibility and long-term maintainability.
- Built ML Pipelines for tire-wear modelling, geo-location tracking, and battery state-of-charge estimation using supervised learning, feature engineering, time series analysis, regression models, model evaluations with automated data preprocessing, training, validation and deployment.
- Designed reusable Python libraries and data-processing modules that standardize workflows and accelerate delivery velocity.
- Conducted structured Python training sessions for 20+ engineering teams, driving measurable adoption of programming-based analytics across Volvo teams.

Senior Systems Engineer · Infosys

April 2018 – October 2021

- Built OCR-based data extraction pipelines for Novartis (pharmaceutical domain), achieving ~80% extraction accuracy on complex multi-format regulatory documents.
- Performed root cause analysis on OCR failures and extraction inconsistencies, implementing enhancements that improved pipeline reliability.
- Automated structured data extraction from unstructured PDFs and scanned images, reducing manual data-entry effort for compliance-critical workflows.
- Designed a PostgreSQL data storage architecture to centralize extracted product information, enabling efficient querying and downstream reporting for QA and business teams.
- Developed Power BI dashboards that provide business users with access to product information and quality metrics, replacing a manual reporting cycle.
- Created monitoring and logging mechanisms to track document processing status, extraction quality, and pipeline failures.

EDUCATION

M.Tech in Data Science & Engineering · BITS Pilani, Hyderabad | March 2020 – August 2022

Coursework: Machine Learning, Deep Learning, Big Data Analytics, Statistical Modelling, NLP and Artificial Intelligence

B.E. in Electronics & Communications · Anna University, Chennai | August 2012 – November 2016

Coursework: Foundation in programming, signal processing, embedded systems, and communication networks